

Assignment 2 DESIGN.pdf:

Description of Program:

This program implements a small number of mathematical functions (ex and px) and uses them to compute the fundamental constants e and π . It also includes a dedicated test harness named mathlib-test that compares implemented functions with that of the math library's. There is also a writeup pdf that presents the analysis of the results.

Files to be included in directory "asgn2" + pseudocode/structure:

- viete.c
 - This contains the implementation of Viète's formula to approximate π and the function to return the number of computed factors.
 - double f (factor that repeats itself) = 2
 - double pi = 1
 - While last term is less than EPSILON
 - f equals to f plus square root of f
 - pi equals pi times f divided by 2
 - pi equals pi times square root of 2 divided by 2
 - pi equals 2 divided by pi
 - return pi
- e.c
 - This contains the implementation of the Taylor series to approximate Euler's number e and the function to return the number of computed terms.
 - sum = 1
 - count = 1

```

while sum is less than Epsilon
    sum = sum + 1/ the factorial of count
return sum

```

- euler.c

- This contains the implementation of Euler's solution used to approximate π and the function to return the number of computed terms.

```

count = 1
sum = 1
while count is less than epsilon
    sum = sum + sum divided by count squared
sum = sum * 6
return square root of sum

```

- madhava.c

- This contains the implementation of the Madhava series to approximate π and the function to return the number of computed terms.

```

pi = 0
While pi less than EPSILON
    pi = pi + (-3) to the power of -k / 2 times k plus 1
pi = pi * square root of 12
Return pi

```

- mathlib-test.c

- This contains the main() function which tests each of your math library functions.

- mathlib.h

- This contains the interface for math library

- `newton.c`
 - This contains the implementation of the square root approximation using Newton's method and the function to return the number of computed iterations.
- `bbp.c`
 - This contains the implementation of the Bailey-Borwein-Plouffe formula to approximate π and the function to return the number of computed terms.
- `WRITEUP.pdf`
 - This document must be a proper PDF.
 - Graphs displaying the difference between the values reported by implemented functions and that of the math library's.
 - Use a UNIX tool to produce graphs. `gnuplot` is recommended.
- `Makefile`
 - `CC = clang` must be specified.
 - `CFLAGS = -Wall -Wextra -Werror -Wpedantic` must be specified.
 - `make` must build the `Mathlib-test` executable, as should `make all` and `make mathlib-test`
 - `make clean` must remove all files that are compiler generated.
 - `make format` should format all your source code, including the header files.
- `README.md`
 - In proper Markdown syntax.
 - Describe how to use your program and Makefile
 - Lists and explains any command-line options that program accepts.
 - Reports any false positives by `scan-build`
 - Notes down any known bugs or errors in this file as well for the graders.
- `DESIGN.pdf`
 - This document must be a proper PDF.

- Describes design and design process for program with enough detail such that a sufficiently knowledgeable programmer would be able to replicate implementation.